Vibe Shuffle: Next-Song Selection from Relative Valence–Arousal Trajectories

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Abstract

Music recommender systems learn durable preferences, but the track that fits can change from one moment to the next. *Vibe Shuffle* examines whether lightweight camera and cardiac observations can inform next-song selection without being presented as a readout of emotion. It expresses each channel as change from a personal reference, collects successive relative Valence–Arousal points during playback, and uses their mean to select the nearest eligible track from the matching region of a fixed catalog. An exploratory within-participant crossover compared this policy with Random selection from the same catalog. Mood fit was higher on average under Vibe Shuffle, while liking changed little; both estimates were uncertain and did not establish a recommendation benefit. Each recommendation can be traced from personal references through multimodal observations to song choice. Testing the policy as a session-aware re-ranking layer requires further validation of the signal mappings and study protocol.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; • **Information systems →** *Recommender systems.*

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, baseline-relative adaptation

1 Motivation

Music choice is contextual. A track that suits exercise may be unwelcome during a quiet break. People use music to regulate mood and activation [11, 25]. Most recommender systems prioritize durable signals such as listening history, favorite artists, similar users, and item features. Those signals characterize usual preference and capture momentary fit only indirectly, a known challenge in contextual music recommendation [12, 26]. A listener may like a song in general without wanting it next.

Explicit mood selection can address that gap, but it also interrupts an otherwise continuous experience. Automatic adaptation offers another route. Its observations, however, are uncertain. Facial behavior depends on pose, lighting, culture, context, and deliberate expression [2, 5, 14]. Cardiac timing varies with respiration, posture, movement, fitness, medication, and sensor quality [15, 23, 27]. Neither channel directly identifies emotion.

Vibe Shuffle uses these uncertain observations only as within-listener changes that may guide the next song. During setup, the listener provides short personal camera and cardiac references. Later observations are expressed as changes from those references. The horizontal axis represents relative Valence evidence; the vertical axis represents relative Arousal evidence. Each point falls into a Tense, Energetic, Melancholic, or Calm recommendation region. These names identify catalog regions, not the listener's emotional state.

The catalog uses song Valence and Energy coordinates. Vibe Shuffle averages the recent trajectory, retains unplayed tracks in its region, and selects the nearest eligible track. Random draws from the same remaining catalog without the trajectory. The adaptive chain is explicit: personal references, relative observations, trajectory, region, and nearest available song.

The study compares this policy with Random selection from the same fixed catalog. Its primary outcome is participant-rated mood fit, analyzed through the paired difference and the uncertainty around it. Liking is a secondary measure of general song preference. The comparison rests on three reproducible elements:

- Personal reference periods center camera and cardiac observations before the relative Valence and Arousal coordinates are calculated.
- Event-driven observations form a trajectory whose mean identifies a catalog region and whose distance ranking yields the nearest unplayed track.
- The exploratory crossover estimates mood-fit and liking differences under matched catalog, exposure, and rating procedures.

Mood-fit estimates favored Vibe Shuffle, whereas liking barely changed; both intervals crossed zero. A service-level evaluation would require validated sensor mappings, complete trial records, preregistration, and a dynamic catalog.

2 Background

Dimensional affect models provide a shared vocabulary for listeners and music. Russell's circumplex represents Valence, from unpleasant to pleasant, and Arousal, from low to high activation [22, 24]. Thayer's account adds complementary categories for energetic and tense activation [28]. Both dimensional and categorical models occur in music-emotion research, and results depend on the selected model, labels, and stimuli [7]. Vibe Shuffle adopts this vocabulary

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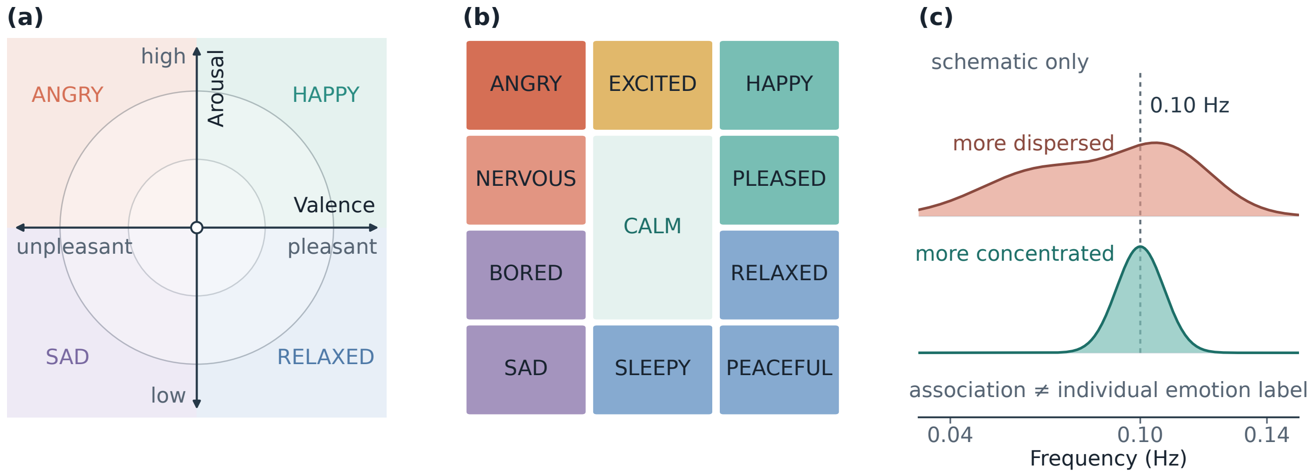

Figure 1: Panels (a)–(b) are original redrawings adapted from Helmholz et al. [9], who operationalized Russell's circumplex [24] and Thayer's mood–arousal model [28]. Panel (c) is an original schematic adapted from the group-level frequency distinction reported by Balaji et al. [4]; it contains no empirical observations.

as a coordinate for recommendation, not as a complete account of experience.

Figure 1 contrasts continuous coordinates, a categorical vocabulary, and HRV-frequency context. The policy outputs neither emotion words nor coherence profiles. Its coordinates encode within-listener change, which does not make facial or cardiac observations emotion-specific.

Songs occupy a parallel Valence–Energy space. Song Valence describes conveyed positivity, whereas Energy describes intensity and activity. Prior interfaces have used these catalog features as counterparts of affective Valence and Arousal [9]. Vibe Shuffle follows that convention while keeping the constructs distinct: musical Energy is not physiological Arousal. Music-emotion coordinates depend on subjective annotation and compress a course that changes over time [1, 30].

Splitting both axes at 0.5 yields four recommendation regions. The listener's relative position selects a region, region membership determines eligibility, and Euclidean distance ranks the eligible songs. The coordinate functions as a controlled retrieval space and makes no claim to ground-truth emotion.

Cardiac timing supplies one source of relative Arousal information. HRV describes beat-to-beat variation, and RMSSD summarizes short-term RR-interval variation [27]. Recording length, respiration, movement, posture, fitness, medication, contact quality, and vendor processing limit interpretation [15, 23]. Vibe Shuffle uses heart rate and RMSSD only as baseline-relative Arousal evidence. Its custom spectral-peakedness field is diagnostic, differs from the group-level coherence measure in Balaji et al. [4], and never affects selection.

3 Related Work

Music can evoke, express, maintain, or regulate affect [11]. Emotion-aware systems use explicit or inferred context to constrain the next track. *Feel the Moosic* maps a listener-selected affect position to Spotify Valence–Energy ranges [9]. Vibe Shuffle instead derives the coordinate from baseline-relative camera and cardiac observations over a listening window.

Janssen et al. learned listener-specific relations between music and physiological response and used them to steer an affective music player toward a goal [10]. Ayata et al. estimated Valence and Arousal from wearable galvanic-skin-response and photoplethysmography measurements before recommending music [3]. Tran et al. matched a facial estimate with nearby tracks in Spotify feature space [29]. These systems demonstrate several routes from personal context to track selection. Their reported evaluations provide no clear evidence that the selected songs improve momentary fit.

The path from observation to selection can hide consequential assumptions. Explanations may support transparency, trust, effectiveness, or decision support [21]; perceived quality also depends on effort, privacy, and context [13]. A reviewable policy should expose what was observed, how observations were summarized, and why one song became eligible.

Vibe Shuffle differs in three connected choices: it centers multimodal observations on a personal reference, retains their movement as an event-driven trajectory, and applies a fixed region-and-distance rule. Random selection uses the same frozen catalog, exposure, and rating procedure. This comparison isolates the policy, while the exploratory data leave its benefit uncertain.

4 Concept Overview

Vibe Shuffle was designed around personal calibration, inspectable decisions, and cautious interpretation. Observations are expressed as changes from the listener's own reference instead of population thresholds. The research record exposes the path from those observations to the selected song, while sensor outputs guide recommendation without being shown as the listener's emotion.

The project evaluates whether tracks match the present session. It does not attempt to regulate mood or assume that an energetic track should calm activation or that positive Valence should repair a negative state. Maintaining, intensifying, and shifting would require separate policies.

The final design combines a personal reference, a trajectory built from multiple samples, a fixed region-and-distance rule, and a Random comparator from the same catalog. Optional inputs and explicit signal-path fields let a session continue when a channel is missing. The repository documents the implemented system and its later methodological audit, but it contains no sketches, interviews, dated prototype history, or qualitative feedback log from which participant-driven design changes could be reconstructed.

Vibe Shuffle follows a six-step loop: acquire personal references, collect camera and cardiac observations during playback, express them as relative Valence–Arousal points, summarize the trajectory, choose the nearest eligible song in the corresponding catalog region, and record the rating and decision trace. Adaptation occurs between tracks, so brief fluctuations cannot interrupt the current song. Figure 2 shows this loop.

The intended use is scientific evaluation of baseline-relative adaptation. A future music service could test the policy as a re-ranking layer above a conventional preference model. The present project neither implements nor validates that deployment.

5 Signal & Pipeline

The multimodal input has two distinct channels. A camera supplies visual and behavioral cues; an optional wearable supplies physiological timing. The camera reference runs for up to 14 seconds and may be skipped. The cardiac reference records 120 seconds of device-produced heart rate and RR intervals. Subsequent observations are centered on these personal values. The references do not define a neutral or true emotional state; they provide local comparison points for the session.

The camera supplies visible evidence primarily for relative Valence; cardiac change supplies relative activation. Both are accessible through commodity interfaces, but each mapping requires separate validation. Figure 2 shows how the relative inputs enter the trajectory and catalog decision.

Once the references are established, each channel contributes to the relative coordinate.

MediaPipe Face Landmarker runs locally and returns facial blendshape activations [17]. A fixed heuristic combines selected smile, cheek, brow, eye, jaw, frown, and mouth cues into positive, negative, and tension-related scores. Samples scheduled at 120 ms intervals are debiased against the reference and smoothed across frames. More positive visible evidence moves relative Valence right; negative or tension-related evidence moves it left. A slowly adapting reference follows longer changes. If no face is detected, Valence returns to the center.

Visible movement and nodding can also add positive-Valence and upward-Arousal evidence. The channel represents change in observable behavior under a fixed rule, not felt emotion. Lighting, pose, occlusion, morphology, culture, deliberate expression, dancing, and repositioning may change the output [2, 5, 14]. The complete coefficients and gates are retained in `src/expressionModel.js`.

Cardiac evidence enters through a compatible Bluetooth Heart Rate Service device, which supplies beats per minute and, when available, RR intervals. The research system has no raw electrocardiogram and cannot inspect the device's beat detector. Implausible or abruptly changing intervals are rejected before variability is calculated. The reference stores median heart rate, the root mean square of successive differences (RMSSD), and robust spread estimates.

During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward Arousal evidence; changes in the opposite direction provide downward evidence. Heart rate receives the larger influence, and RMSSD is damped when the indicators disagree. At least 20 accepted RR intervals are required for the full estimate. RMSSD is a recognized short-term variability summary, but duration, respiration, posture, movement, fitness, medication, contact quality, and vendor processing affect its interpretation [15, 23, 27]. Agreement between 120-second and longer RMSSD measurements has been reported under controlled resting conditions, not short listening windows with movement [19].

The export also records SDNN and a custom spectral-peakedness field called physiology coherence. It differs from the coherence measure associated with post-session emotion labels in large-scale biofeedback data [4]. It is diagnostic only and does not affect song selection.

The channel-specific values enter a shared pipeline for acquisition, plausibility filtering, personal-reference centering, smoothing, quality assessment, and coordinate fusion. Usable cardiac evidence sets the Arousal base, while camera movement can add upward evidence. Camera movement supplies an upward-only fallback when cardiac data are unavailable. If neither channel is usable, the coordinate returns to the central reference point.

Raw camera frames, facial landmarks, and Bluetooth notifications remain in browser memory. A locally downloaded CSV contains derived summaries, reference status, signal source, quality flags, timing, selected-song metadata, and ratings. It excludes frames, landmarks, raw packets, waveforms, complete trajectories, personal facial-reference values, and the session seed. This is a data-minimizing local export, not a formally privacy-preserving system. Spotify playback, MediaPipe assets, and hosting still require network traffic.

6 Inference & Adaptation

During confirmed playback, a change in the fused coordinate appends a point to the trial trajectory

$$\mathcal{T} = \{(V_1, A_1), \ldots, (V_m, A_m)\}.$$

At rating time, the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m} \sum_{i=1}^{m} (V_i, A_i)$$

becomes the target for the next Vibe choice. Aggregation reduces dependence on one observation but is event-weighted, because points are added when the coordinate changes rather than at a fixed rate. The export records sample count and listening duration for review.

Splitting both axes at 0.5 produces four catalog regions: Energetic, Calm, Tense, and Melancholic. These names identify Valence–Energy partitions; they do not classify the participant. Listener

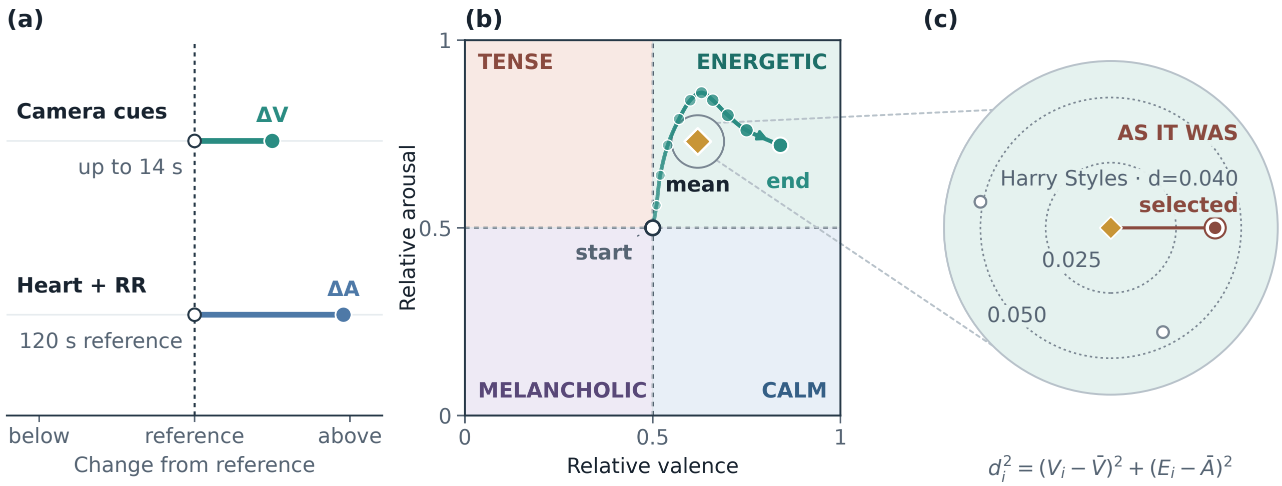

Figure 2: Panel (a) shows signed camera and cardiac changes measured against personal references. Panel (b) summarizes the trajectory between its open start and filled endpoint using the arithmetic mean (gold diamond). In panel (c), the nearest catalog track is *As It Was* by Harry Styles, which the policy therefore selects. Its Euclidean distance is shown beside the track; two unlabeled open points indicate nearby alternatives, and the equivalent squared-distance expression appears below the panel. Panels (a)–(b) are schematic, whereas panel (c) uses catalog-derived coordinates and distance. Coherence is logged but not used for selection.

Arousal and song Energy are operational counterparts for retrieval, not equivalent constructs [9, 30].

The frozen catalog contains 100 unique Spotify tracks, with 25 in each region. Its retained Valence and Energy coordinates originate from a public historical dataset [18]. The repository preserves the track records but not the download date or a script reproducing their curation. The preserved catalog is a frozen stimulus set, not a fresh derivation.

Selection applies the same exclusion rule in both conditions: the current track and every previously played track are removed. Vibe then keeps unplayed tracks in the target region and selects the smallest Euclidean distance:

$$s^* = \arg\min_{s \in Q(\bar{V},\bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

A session seed resolves exact ties. If the region is exhausted, the policy searches the complete unplayed pool. Random uses the same pool and exclusions but ranks candidates by a session-seeded pseudorandom score. Catalog, playback path, exposure count, and rating questions remain constant, isolating the selection rule more closely than a comparison between different interfaces.

Because sensing is optional, a Vibe decision can use camera evidence, cardiac evidence, both, or neither. The retained summaries do not report these frequencies; the first track may also precede a completed facial reference.

Adaptation changes only the next-song ranking. The completed trajectory produces a target, region, eligible set, distance ranking, and selected track. The local record exposes the final coordinate and decision, although it cannot replay every sensing step.

In a future service, a long-term preference model could first generate a suitable candidate pool. A confidence-gated Vibe policy could then adjust the ordering for the current session while preserving an override and a visible explanation. That deployment model has not been evaluated.

The decision record contains the personal reference, relative coordinate, trajectory mean, catalog region, eligible set, distances, and chosen track. In a service, these fields could support an explanation, suppress adaptation under weak signals, and preserve immediate preference-only selection.

7 User Study & Reflection

The exploratory study used a within-participant crossover with five-track Random and Vibe blocks drawn from the same frozen catalog. Two protocols reversed block order; the current interface suggests an assignment from the participant number and masks condition names. Administration can still reveal the condition, and the study was not double blind. Reversing order reduces but does not remove practice, fatigue, or carryover risks [8].

Each track plays for up to 60 seconds, with an option to rate earlier. After each track, participants answer *How much do you like this song?* and *How well did it fit your current mood?* on seven-point scales. They also select Energetic, Calm, Tense, Melancholic, or Neutral as a categorical mood report. Mood fit measures the proposed contextual match; liking distinguishes that judgment from general song preference. Listening duration, early rating, condition order, and the current implementation's selection fields are exported.

In the current protocol, the experimenter selects a masked order and pseudonymous participant number after Spotify authentication. The participant completes available references, hears and rates five tracks per condition, and downloads the local study file. Vibe uses the preceding listening window; Random ignores it. Because the software revision used for the pilot was not retained, the procedure described here corresponds to the present implementation and may differ from the sessions summarized in the archive.

The retained data contain 15 participant/session rows under generic identifiers, each with one five-trial condition mean for mood fit and liking. Trial rows, participant characteristics, study administration, hardware, and a code revision tied to collection are unavailable. These records cannot reconstruct trial completeness, exclusions, within-session variation, or signal-path frequencies.

Paired differences are Vibe minus Random, and the participant/session five-trial mean is the unit of analysis. We report condition means and standard deviations, paired mean differences, two-sided 95% confidence intervals, standardized paired effects, and the supplied directional paired tests. Confidence intervals use the paired t distribution with 14 degrees of freedom. Figure 3 adds 20,000 fixed-seed bootstrap resamples to show the mean distribution; its interval bars remain t-based.

The archived means are rounded to one decimal place and reproduce the supplied condition summaries and paired t results to rounding. Exact signed-rank values depend on unavailable precision and tie handling, so we retain the supplied values. No dated preregistration or confirmatory threshold was available. Estimation and two-sided uncertainty are primary; the one-sided tests are reported descriptively [6, 20]. Mood fit averaged 4.71 under Vibe and 4.15 under Random (Table 1). The paired difference was +0.56 rating points, with a 95% confidence interval from −0.21 to 1.33 and a standardized paired effect of $d_z = 0.40$. Eight participant/session means were higher under Vibe, one was tied, and six were lower. The retained directional paired test yielded $t(14) = 1.56$, $p = .070$; the supplied signed-rank sensitivity check yielded $W^+ = 74.0$, $p = .088$. The interval includes effects on both sides of zero.

Liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09, with a 95% confidence interval from −0.52 to 0.71 and $d_z = 0.08$. The directional tests yielded $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. No equivalence margin was specified, so the near-zero estimate does not establish equivalence.

The data leave both effects unresolved. Mood-fit ratings were higher on average under Vibe Shuffle, whereas liking changed little, but neither estimate excludes zero or establishes a recommendation benefit.

Participant/session differences span both directions, with several exceeding one rating point. The retained means cannot distinguish familiarity, context, order, signal availability, or differences in the value of matching; one average should not be treated as a universal personalization effect.

Personal references provide a local comparison point, but the facial and cardiac mappings remain unvalidated. A trajectory makes the decision easier to inspect, but its arithmetic mean gives more weight to periods with more coordinate changes. The matched Random condition controls catalog size, playback path, exposure count, and rating procedure; selected songs can still differ in artist, genre, language, familiarity, and popularity.

The current specification describes sensor outputs as baseline-relative evidence, keeps diagnostic coherence outside the selection rule, and records the target, region, distance, signal source, and quality flags. These reporting changes do not alter the historical data, and no qualitative feedback archive supports claims about participant-driven redesign.

Sixty-three deterministic tests cover catalog balance, export quality flags, facial rules, RR filtering, baseline-relative cardiac behavior, and song selection. They validate the present modules, not sensor accuracy, playback, the end-to-end study flow, or the pilot software revision.

The facial and cardiac mappings are fixed heuristics without independent validation. Movement may alter camera evidence, raise heart rate, and degrade RR quality at the same time, allowing one behavior to influence both axes. Short HRV windows and unmeasured respiration further constrain interpretation.

The small exploratory crossover and incomplete archive also prevent trial-level, order, carryover, missing-data, and subgroup analyses. The retained files contain no recruitment, ethics, hardware, or trial-level records, so those parts of the study cannot be reconstructed. One-sided tests should not be treated as confirmatory without preregistration. The first adaptive choice may precede completed references, sensing is optional, and the historical summaries do not identify the signal path behind each choice.

The frozen catalog improves experimental control but limits ecological validity. The conditions share a catalog, not identical tracks. The system also uses event-weighted rather than time-weighted aggregation. At the exact no-signal center, current code paths disagree over whether the boundary belongs to Calm or Energetic. That edge case should be unified before another study.

These limitations set the priorities for the next evaluation. It should begin by testing the frozen mappings under controlled changes in expression, lighting, posture, movement, and respiration. An instrumented pilot should require completed references, timing and missing-data flags, and all ten trial records. Camera-only, cardiac-only, fused, and Random conditions would identify the useful evidence path; time-weighted and event-weighted trajectory summaries should be compared directly.

Before a larger crossover, preregister the smallest relevant mood-fit effect, sample size, exclusions, paired and order analyses, and signal-quality rules [16]. A dynamic-catalog study should record availability, playback failures, coverage, revisions, explanations, overrides, and privacy expectations. Recording these factors would distinguish measurement failures from recommendation failures and show whether session-aware re-ranking adds value above a preference model.

8 Conclusion & Future Work

Vibe Shuffle turns within-listener camera and cardiac changes into next-song decisions through personal references, a listening trajectory, and an explicit region-and-distance rule. The resulting chain from observation to selected track can be inspected without assigning the listener an emotion label.

The crossover yielded a higher average mood-fit rating under Vibe Shuffle and almost no change in liking, but both intervals crossed zero. The available records cannot explain the between-participant differences, so the policy's effect remains unresolved.

If later evidence supports the policy, session-aware re-ranking could be tested for exercise, focus, relaxation, games, or accessibility settings. Real-world use would require visible uncertainty, an immediate opt-out, data-minimizing processing, and adaptation subordinate to explicit choices.

Table 1: The values are exploratory five-trial means from 15 participant/session records. Confidence intervals are two-sided; the retained tests use one-sided Vibe-over-Random alternatives.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]	$t(14)$, p	W^+, p	d_z
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]	1.56, .070	74.0, .088	0.40
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]	0.32, .375	58.5, .353	0.08

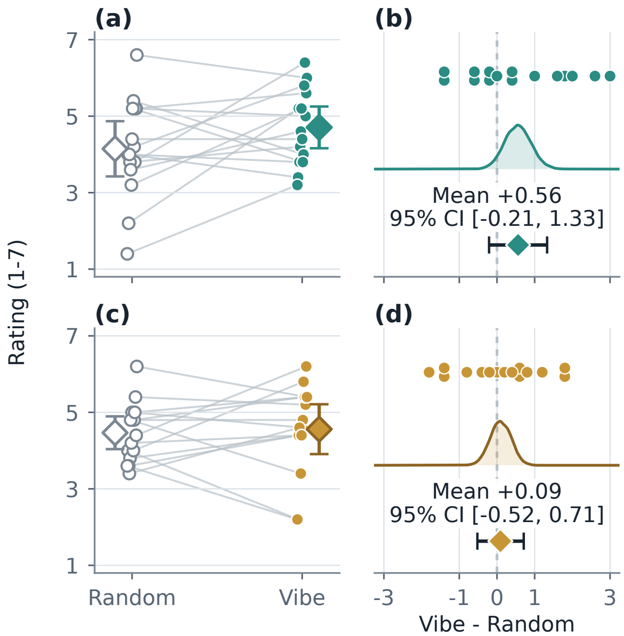

Figure 3: Panels (a)–(b) show mood fit, while panels (c)–(d) show liking. The left panels connect Random and Vibe five-trial means and summarize condition means. The right panels show all paired differences, fixed-seed bootstrap mean distributions, and reported paired means with t-based 95% CIs. Both paired intervals cross zero.

The next study must validate the signal mappings, retain complete trial records, preregister its analysis, and evaluate a dynamic candidate pool. Only then can session-aware re-ranking be judged against preference-based recommendation in a real music service. Vibe Shuffle supplies the inspectable mechanism; the next experiment must determine whether it improves what listeners hear.

9 Contribution Statement

Markus Baumann, Felix Hajuj, and Miriam Janner jointly contributed to the project concept, study execution, interpretation, and review of the report. Markus Baumann additionally implemented and tested the archived repository version, reproduced the participant-level statistics, and prepared the visual analysis. The available records do not preserve a finer-grained division of writing responsibilities.

Open Science and Transparency

The project repository contains source code, the frozen catalog, tests, figure scripts, participant/session means, and result-provenance records: https://github.com/eybmits/vibe-shuffle. It contains no raw trial rows, sensor streams, recruitment records, administration logs, or code revision tied to the pilot sessions. The computational tests verify the reported source tree, not the historical data-collection environment.

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